Cookie Cats A/B Testing Analysis: Evaluating Gate Placement on Player Retention

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Abstract

This report presents an A/B testing analysis of the Cookie Cats mobile game, examining whether gate placement at level 30 (*gate_30*) versus level 40 (*gate_40*) affects player retention across 90,189 players. Both Frequentist and Bayesian approaches are applied to binary retention metrics (*retention_1*, *retention_7*) and a continuous game rounds metric (*sum_gamerounds*).

Frequentist results show a statistically significant difference in 7-day retention ($\chi^2(1) = 9.959$, $p = 0.0016$), while 1-day retention shows no significant difference ($p = 0.0755$). Bayesian posterior analysis confirms a 99.92% probability that *gate_30* yields higher 7-day retention, with an estimated advantage of 0.31%–1.33%. All effect sizes remain negligible, though the consistency across both methods supports retaining the gate at level 30.

1 Introduction

Player retention is a critical metric in the mobile gaming industry, as retaining existing players is substantially more cost-effective than acquiring new ones. This report presents an A/B testing analysis of the Cookie Cats mobile game, examining whether gate placement at level 30 (*gate_30*) versus level 40 (*gate_40*) affects player retention.

The dataset contains both binary outcomes (*retention_1*, *retention_7*) and a continuous metric (*sum_gamerounds*), requiring different statistical approaches. A Frequentist analysis is first applied, using the Chi-square test for binary outcomes and the Mann-Whitney U test for the right-skewed continuous metric. While Frequentist testing confirms whether a difference is statistically significant, it cannot quantify how probable that difference is. A Bayesian approach is therefore applied using conjugate models to provide posterior probabilities and credible intervals for a more interpretable conclusion.

2 Data

2.1 Dataset Overview

The dataset contains 90,189 player records sourced from Kaggle, collected from a randomized A/B experiment conducted on the mobile puzzle game Cookie Cats. Players were randomly assigned to one of two experimental groups upon installation: *gate_30* (control, $n = 44{,}700$) and *gate_40* (treatment, $n = 45{,}489$). The two groups are well-balanced in size, with no missing values detected across all variables. The primary outcome variables are *retention_1* and *retention_7*, both binary indicators representing whether a player returned to the game 1 day and 7 days after installation, respectively.

2.2 Feature Description

The dataset contains 5 variables: 1 identifier (*userid*, excluded from analysis), 1 grouping variable (*version*), 1 behavioral metric (*sum_gamerounds*), and 2 binary outcome variables (*retention_1*, *retention_7*). *version* indicates the experimental group assigned to each player, either *gate_30* (control) or *gate_40* (treatment). *sum_gamerounds* records the total number of game rounds played within the first 14 days after installation. *retention_1* and *retention_7* are logical variables indicating whether a player returned to the game 1 day and 7 days after

installation, respectively.

3 Exploratory Data Analysis

3.1 Summary Statistics

The dataset contains 90,189 complete observations with no missing values detected. *sum_gamerounds* ranges from 0 to 49,854 with a mean of 51.87 and a median of 16, indicating a heavily right-skewed distribution. The two experimental groups are well-balanced: *gate_30* contains 44,700 players (49.6%) and *gate_40* contains 45,489 players (50.4%). Overall 1-day retention rate is 44.5% and 7-day retention rate is 18.6%.

3.2 Feature Distribution by Version

Figure 1 shows the retention rates for both experimental groups. *gate_30* and *gate_40* show similar 1-day retention rates (approximately 44.8% vs. 44.2%), while *gate_30* shows a slightly higher 7-day retention rate (19.0% vs. 18.2%).

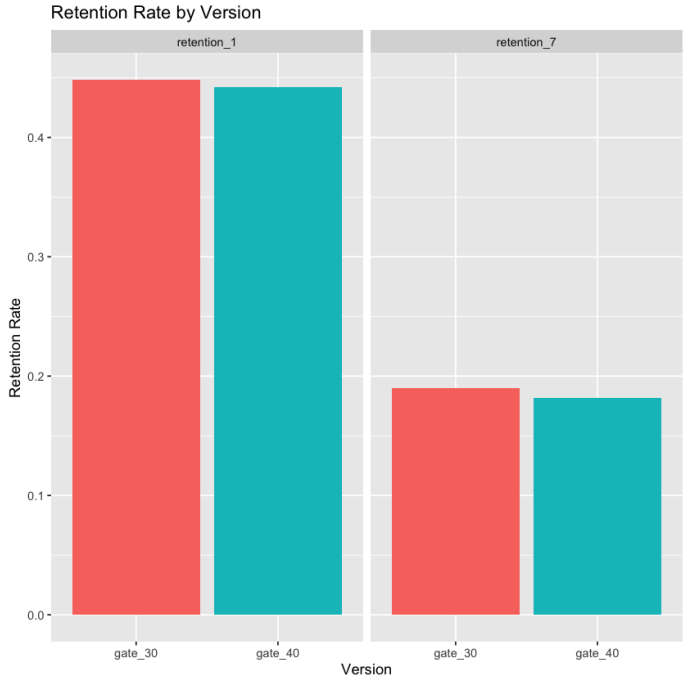

Figure 1: Retention Rate by Version

3.3 Outlier Detection

To assess the presence of extreme values in *sum_gamerounds*, the IQR method was applied. Figure 2 visualizes the IQR $\times$ 3 outlier threshold across both groups on a log scale. A total of 5,704 observations (6.32%) exceed the upper bound. Since *sum_gamerounds* is a secondary metric and the definition of a game round is ambiguous, outliers are retained in the analysis.

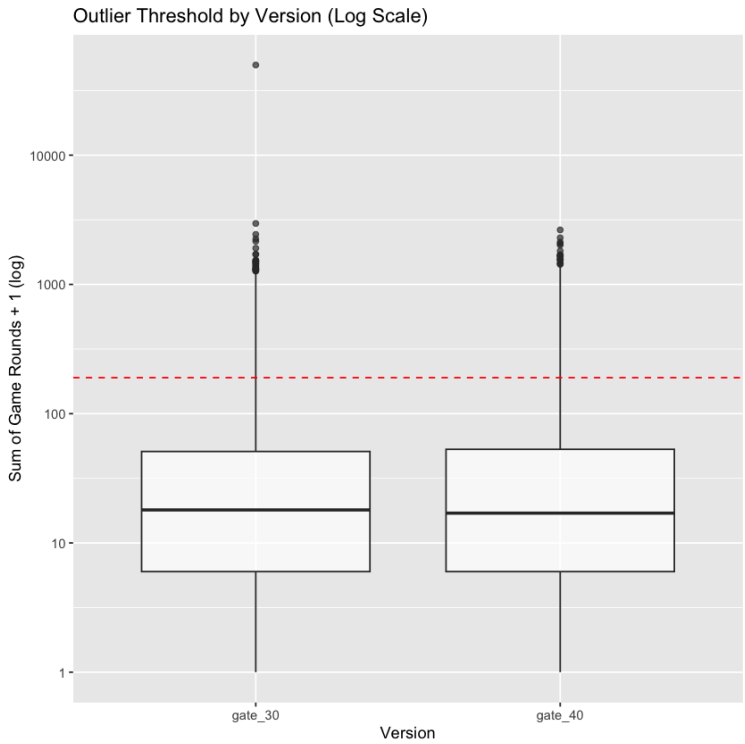

Figure 2: Game Rounds by Version (Log Scale). The dashed red line indicates the IQR $\times$ 3 outlier threshold.

4 Methodology

4.1 Frequentist Approach

Chi-Square Test The Chi-square test of independence was applied to evaluate whether gate placement is associated with binary retention outcomes (*retention_1*, *retention_7*). The null hypothesis states that retention is independent of version assignment.

Mann-Whitney U Test The Mann-Whitney U test was applied to compare *sum_gamerounds* between the two groups. As a non-parametric test, it does not assume normality and is therefore appropriate for skewed count data.

4.2 Bayesian Approach

Beta-Binomial Model For binary retention outcomes, a Beta-Binomial conjugate model was applied with an uninformative prior Beta$(1, 1)$, yielding the posterior:

$$p \mid \text{data} \sim \text{Beta}(\alpha + \text{successes},\ \beta + \text{failures})$$

Gamma-Poisson Model For *sum_gamerounds*, a Gamma-Poisson conjugate model was applied with prior Gamma$(1, 1)$, yielding the posterior:

$$\lambda \mid \text{data} \sim \text{Gamma}\Big(\alpha + \sum x_i,\ \beta + n\Big)$$

4.3 Evaluation Metrics

Frequentist tests are evaluated using p-value and effect size: Cramér's V for Chi-square tests and rank-biserial r for the Mann-Whitney U test.

For Bayesian models, Monte Carlo sampling ($n = 100{,}000$) was used to estimate the following metrics:

- $P(\textit{gate_30} > \textit{gate_40})$: posterior probability that *gate_30* outperforms *gate_40*
- **95% Credible Interval**: the range of plausible differences with 95% posterior probability

5 Results

5.1 Frequentist Results

Table 1 presents the Chi-square test results for both retention metrics. *retention_7* shows a statistically significant association with gate placement ($\chi^2(1) = 9.959$, $p = 0.0016$), while

retention_1 does not reach significance ($\chi^2(1) = 3.159$, $p = 0.0755$). However, both Cramér's V values are below 0.02, indicating a negligible practical effect.

Metric	χ^2	**df**	**p-value**	**Cramér's V**
retention_1	3.159	1	0.0755	0.0059
retention_7	9.959	1	0.0016	0.0105

Table 1: Chi-square test results for retention metrics

Figures 3 and 4 show the retention rates with 95% confidence intervals for each group. For *retention_1*, *gate_30* achieves 44.8% and *gate_40* achieves 44.2%, with overlapping intervals indicating no reliable difference. For *retention_7*, *gate_30* achieves 19.0% and *gate_40* achieves 18.2%, with non-overlapping intervals confirming a statistically significant difference.

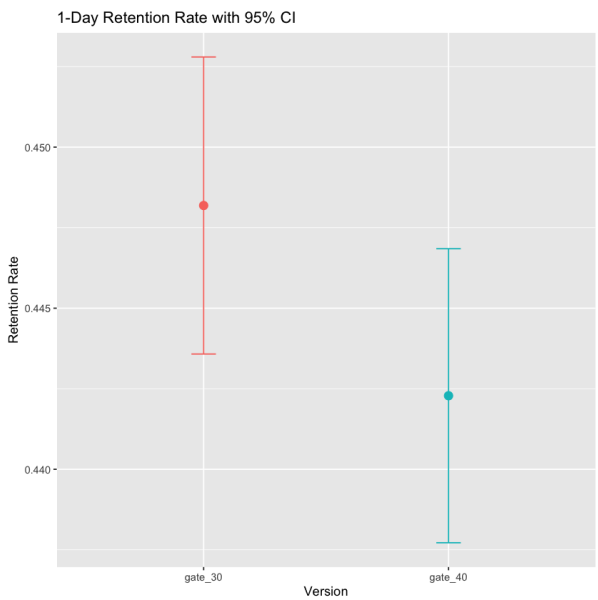

Figure 3: 1-Day Retention Rate (95% CI)

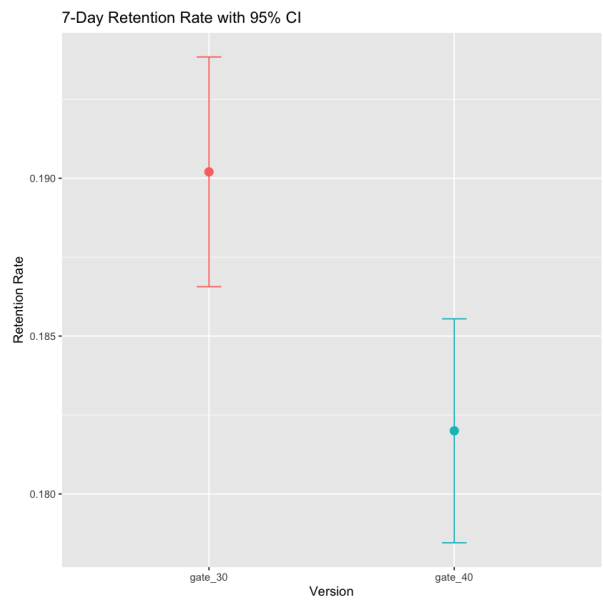

Figure 4: 7-Day Retention Rate (95% CI)

For *sum_gamerounds*, the Mann-Whitney U test yields $W = 1{,}024{,}331{,}250$, $p = 0.0502$, with a rank-biserial correlation of $r = -0.0075$, indicating no significant difference and a negligible effect size between the two groups.

Metric	**W**	**p-value**	**Rank-Biserial r**
sum_gamerounds	1,024,331,250	0.0502	−0.0075

Table 2: Mann-Whitney U test results for game rounds

5.2 Bayesian Results

Table 3 presents the Bayesian posterior results for all three metrics. For *retention_1*, the probability is 96.19% with a credible interval of $[-0.0006, 0.0124]$, which includes zero, suggesting less certainty. For *retention_7*, there is a 99.92% posterior probability that *gate_30* outperforms *gate_40*, with a 95% credible interval of $[0.0031, 0.0133]$, indicating *gate_30* retains 0.31%–1.33% more players after 7 days. For *sum_gamerounds*, *gate_30* players accumulate on average 1.06–1.25 more game rounds over the 14-day observation period with 100% posterior probability.

Metric	$P(\boldsymbol{gate_30} > \boldsymbol{gate_40})$	**CI Lower**	**CI Upper**
retention_1	96.19%	−0.0006	0.0124
retention_7	99.92%	0.0031	0.0133
sum_gamerounds	100.00%	1.0631	1.2519

Table 3: Bayesian posterior results

Figures 5 and 6 show the posterior distributions of the retention rates for both groups. The overlapping distributions in *retention_1* reflect greater uncertainty, while the narrow, well-separated distributions in *retention_7* confirm high certainty in the superiority of *gate_30*.

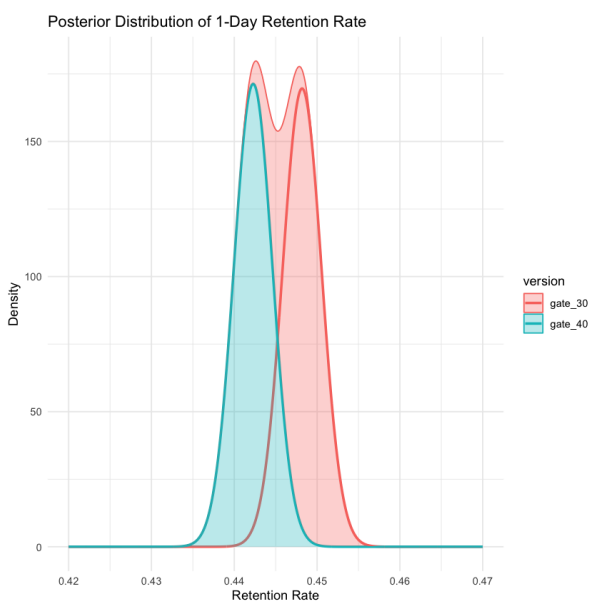

Figure 5: Posterior Distribution (1-Day)

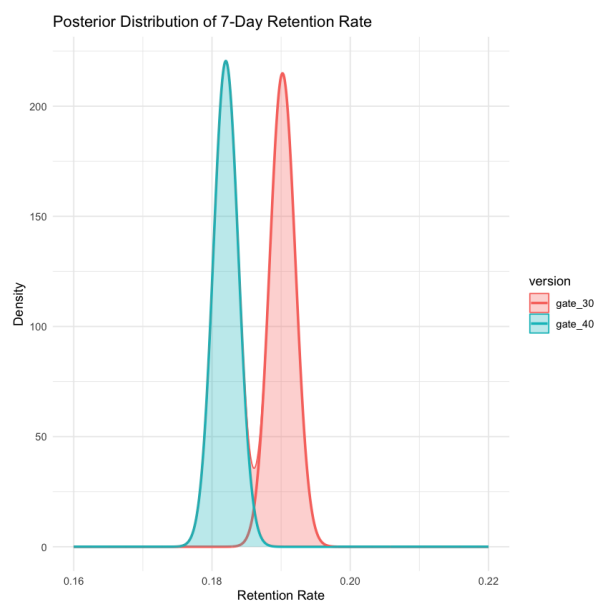

Figure 6: Posterior Distribution (7-Day)

Figure 7 shows the posterior distribution of the average game rounds rate under the Gamma-

Poisson model. The two distributions are well-separated, reflecting high certainty that *gate_30* players play more rounds on average.

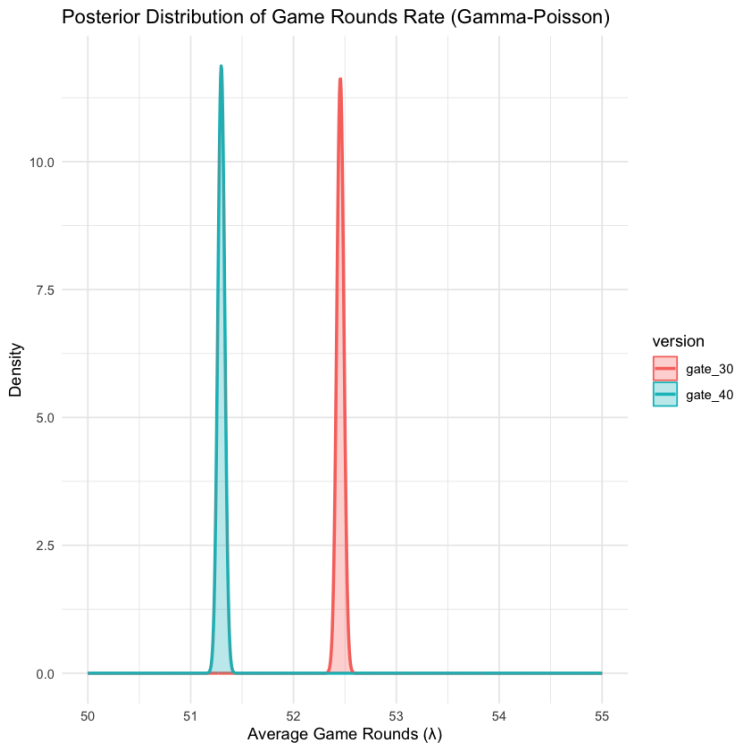

Figure 7: Posterior Distribution of Game Rounds Rate (Gamma-Poisson)

5.3 Overall Comparison

Both Frequentist and Bayesian approaches consistently indicate that *gate_30* outperforms *gate_40* across all three metrics. The Frequentist approach confirms a statistically significant difference in *retention_7* ($p = 0.0016$), while *retention_1* and *sum_gamerounds* do not reach significance. The Bayesian approach provides a more interpretable conclusion: *gate_30* is superior in 7-day retention with 99.92% posterior probability, and all three metrics show posterior probabilities above 96%. Notably, all effect sizes remain negligible, suggesting the practical impact is limited despite statistical significance.

6 Discussion & Conclusion

6.1 Key Findings

Both Frequentist and Bayesian analyses consistently indicate that *gate_30* outperforms *gate_40*. The effect is most pronounced in *retention_7*, where both methods converge on the same conclusion with high certainty.

A notable observation is that *retention_1* shows no reliable difference between the two groups, suggesting that the gate placement effect takes time to manifest. Players are unlikely to have reached the gate within the first day, making 1-day retention an insufficient metric for evaluating gate effects.

Despite statistical significance in *retention_7*, all effect sizes remain negligible. This is a common pattern in large-sample A/B tests, where even trivial differences can reach significance.

6.2 Business Recommendations

Based on the analysis, **retaining the gate at level 30 is recommended**. Placing the gate earlier forces players to stop before reaching full satisfaction, leaving an unfinished desire that motivates them to return. This aligns with the theory of *hedonic adaptation*: enforced breaks preserve the novelty of the game and sustain long-term engagement.

In contrast, *gate_40* allows players to play longer before being interrupted. By the time the gate appears, players have already experienced sufficient satisfaction, reducing their motivation to return. The result is a measurable decline in 7-day retention.

While the retention advantage of *gate_30* is statistically significant, the effect size remains small. At scale, however, even a marginal improvement in retention carries meaningful implications for revenue and player lifetime value.

6.3 Limitations

Dataset Limitations

The definition of *sum_gamerounds* is ambiguous, as it is unclear whether one round corresponds to one level attempt or one completed level. This limits the interpretability of the

Gamma-Poisson results and the outlier analysis. Additionally, the dataset does not include contextual variables such as player demographics, device type, or prior gaming experience, which may confound the relationship between gate placement and retention.

Evaluation Metrics

The analysis relies solely on *retention_1* and *retention_7* as outcome metrics. While 7-day retention captures the gate effect more reliably, a single time point may not fully represent long-term player behavior. Furthermore, the confidence interval plots suggest that the observed differences, while statistically detectable, are narrow in practical terms. Additional metrics such as 14-day or 30-day retention, or revenue per player, could provide a more comprehensive evaluation.

References

[1] Mobile Games (2024). *Cookie Cats* [Dataset]. Kaggle. `https://www.kaggle.com/datasets/yufengsui/mobile-games-ab-testing`